

P5: is the growth series F_T of W_T D -finite? A reduction, not a resolution

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Labels: PROVED means a complete argument, given the inputs it cites. VERIFIED means numerical/computational evidence with the precision or range stated (no error control unless said otherwise). HEURISTIC means an argument without a closed error bound, offered as an informed guess. Notation follows `paper/journal/paper1.tex` (“paper1”) for W_T and `paper/journal/paper2a.tex` (“paper2a”) for U, V and the travel operator.

Summary of outcome

This note does *not* resolve Conjecture `conj:WT-nonDfinite` of paper1 (that F_T , the growth series of the rational right-triangle group W_T , is not D -finite). What it does:

- States precisely what is already PROVED (Theorem `thm:WT-lowcomplexity`: no relation of complexity $\leq d$ in explicit finite boxes) and what is only VERIFIED/HEURISTIC (the linear carry-width growth, the non-transfer of the travel-pole mechanism), reading paper1 §8.7–8.9 in full.
- Isolates the exact mechanism that proves non- D -finiteness for U, V in paper2a (Theorem `thm:nonDfinite`, via Lemma `lem:odeposes`) and states, as a genuinely new but easy PROVED lemma (§4), the precise sufficient condition under which the same mechanism would give non- D -finiteness of F_T : a meromorphic continuation of F_T to a domain containing infinitely many of its poles. This is an immediate corollary of the already-proved domain-general Lemma `lem:odeposes`; the content is the reduction, not new analytic input.
- Explains precisely why that hypothesis is not currently available for F_T (§5): unlike U, V, G_0^T , F_T has no known continuation past its radius of convergence at all, so the sufficient condition of §4 cannot even be tested for F_T with tools now in the repository.
- Records the FKS/Akiyama–Frougny analogy in the precise form paper1 already uses it (§6), and states as HEURISTIC (paper1’s own word) why the rational-base transfer picture is expected to fail for W_T : the numeration base ζ_T has $|\zeta_T| = 1$, so nothing contracts the carry, and the carry-width data computed by `wt_carry` shows linear growth in the sampled range, not the boundedness that drives the BS(p, q), $p \mid q$ rationality proof or the Akiyama–Frougny–Sakarovitch automaticity results.
- States the remaining gap as three independently missing ingredients (§5), none of which is closed here.

The row stays OPEN. No new computation was run for this note beyond re-deriving already-published numbers from the existing `wt_carry/wt_growth` output files, which were re-read but not regenerated (see §7).

1 What W_T and F_T are

Fix a right triangle T with positive, unequal rational legs, encoded (paper1 §8) by coprime integers (c, e) with $0 < e < 2c$ via the minimal polynomial $\mu_T = ct^2 - et + c$ of the rotation number ζ_T (a primitive root of μ_T , $|\zeta_T| = 1$, $\zeta_T \notin \mathbb{R}$). The group

$$W_T = \langle R_0, R_1, R_2 \rangle \leq \text{Isom}^{(2)}$$

is the image of the universal reflection group W_{gen} under the representation ρ_T specializing the generic rotation parameter to ζ_T (paper1, §2283–2340: W_T is dense in $\text{Isom}^{(2)}$, linear over $\mathbb{Q}$, solvable of derived length exactly 3, not finitely presented, quasi-isometric to ${}^2\times$ a horocyclic product of Baumslag–Solitar-type spaces). Its *growth series* with respect to the standard generating set $\{R_0, R_1, R_2\}$ is

$$F_T(x) = \sum_{d \geq 0} u_d^T x^d, \quad u_d^T = \#\{g \in W_T : \ell_T(g) = d\},$$

paper1 line 2548. By Theorem `thm:universality-sharp` the layers u_d^T agree with the T -independent *universal* layers u_d (of W_{gen} /its associated Elt-model) up to a shape-dependent onset depth $n_T \geq c_T$ (Corollary `cor:kernel-classification-restate`), and only deviate from them beyond that depth.

The D -finiteness question (P5). F_T is D -finite (holonomic) if it satisfies a linear ODE $\sum_{i=0}^r Q_i(x) F_T^{(i)}(x) = 0$ with polynomial coefficients Q_i , $Q_r \neq 0$; equivalently (Stanley, *Enumerative Combinatorics* II, §6.4 — cited as in paper1/paper2a, not independently re-verified in this note) (u_d^T) is P-recursive. Conjecture `conj:WT-nonDfinite` (paper1, line 2734) asserts F_T is *not* D -finite, for every such T , and likewise for the “toy groups” $\mathbb{Z}[\zeta_T^{\pm 1}] \rtimes_{\zeta_T} \mathbb{Z}$ with generators $\{1, \text{shift}\}$.

2 What is already proved: a computer-assisted finite-order exclusion

computer-assisted) Theorem `thm:WT-lowcomplexity` (paper1, line 2564): for every such T , F_T satisfies no algebraic relation $\sum_i P_i F_T^i = 0$ with $(k, \max \deg P_i)$ at most $(1, 14), (2, 9), (3, 6), (4, 5), (5, 4), (6, 3)$, and (u_d^T) satisfies no P-recursion with $(r, \max \deg Q_i)$ at most $(1, 14), \dots, (11, 0)$ (paper1 line 2572–2574). The proof uses only the 33 shape-independent universal terms $u_0, \dots, u_{32}$ (certified through u_{22} by a `native_decide` Lean computation, tier (ii) trust; through u_{32} by an exact Rust BFS plus modular-plus-exact certificate, tier (iii) trust — see paper1 §8.6/App. `sec:reproduce`) and full-rank-mod- $(2^{61}-1)$ linear algebra in `code/zeta_probe/wt_growth/guess` which is exact in the direction used (full rank mod p implies full rank over $\mathbb{Q}$) but is a script, not formalized.

PROVED, conditional) Theorem `thm:WT-blowup` (paper1, line 2625): *if* the universal series U is not D -finite (paper2a’s Theorem `thm:nonDfinite`, unconditional there — see §3), then for every complexity bound d only finitely many similarity classes of T have F_T of complexity $\leq d$. The proof is

a clean finite-dimensional nested-kernel argument and is ineffective (ends at line 2670: “The proof is ineffective: it gives no value of $N(d)$ ”).

positive, PROVED) Remark **rem:BS-calibration** (paper1, line 2604): the structurally similar Baumslag–Solitar group $BS(1, 2)$ *does* have a rational growth series (Collins–Edjvet–Gill, cited), of type $(14, 13)$, invisible to a finite-horizon search until ≥ 29 terms are available. This is why Theorem **thm:WT-lowcomplexity**, a finite-order exclusion, cannot by itself be read as evidence for the conjecture: a rational F_T of moderate type would currently be undetected.

None of the above proves Conjecture **conj:WT-nonDfinite**. It proves an exclusion of relations in explicit finite boxes, and a conditional statement whose hypothesis is exactly (an instance of) the conjecture’s cousin for the universal series U — itself proved unconditionally in paper2a, but by a mechanism (§3) that paper1 explicitly records as not transferring to F_T (paper1, line 2766: “does not appear to transfer”).

3 The mechanism that works for U, V, G_0^T

Theorem **thm:nonDfinite** of paper2a proves U, V , and the travel resolvent G_0^T are not D -finite, via:

- (a) a meromorphic continuation of the series to the open unit disc $D = \{|x| < 1\}$ (for G_0^T , unconditionally, from the Fredholm-determinant representation of the travel operator $\mathcal{T} = D_e K$, $K = [q^{\max(s, s')}]_{s, s' \geq 0}$; for U, V , standing on Theorems **thm:model** and **thm:RJ**);
- (b) infinitely many poles of that continuation inside D (for G_0^T unconditional — every q_m in q , every $\pm x_m$ in x ; for U, V , at every x_m resp. $\pm x_m$, from Theorems **thm:V**, **thm:U** and Propositions **prop:Vminusx**, **prop:minusx**);
- (c) Lemma **lem:odeposes** (paper2a, line 1468), a fully general and elementary fact: if f is meromorphic on a domain Ω and solves $\sum_{j \leq r} a_j f^{(j)} = b$ off its poles, with $a_0, \dots, a_r, b$ holomorphic on Ω and $a_r \not\equiv 0$, then every pole of f in Ω is a zero of a_r . Since a_r is holomorphic and not identically zero, its zero set in any relatively compact subdomain is finite. Infinitely many poles in D (from (a),(b)) then contradicts a would-be holomorphic-coefficient ODE on a neighbourhood of $\overline{D}$, hence the coefficients cannot be polynomial either (a polynomial is holomorphic everywhere).

This is the only mechanism the repository currently has for proving non- D -finiteness of a series of this general shape. Any attempt to prove Conjecture **conj:WT-nonDfinite** by the same route needs (a) and (b) to hold for F_T itself (or for some series determining F_T , e.g. the toy groups’ growth series).

4 A trivial but honest reduction

Lemma 1 (Reduction, PROVED — immediate from Lemma **lem:odeposes**). *Fix T . Suppose F_T (equivalently, its analytic continuation from a neighbourhood of 0, if one exists) extends to a meromorphic function on some domain $\Omega \subseteq \mathbb{C}$ containing 0, and that this continuation has infinitely many poles lying in some relatively compact subset of Ω . Then F_T is not D -finite: it satisfies no linear ODE with coefficients holomorphic on Ω (a fortiori none with polynomial coefficients, since a nonzero polynomial has finitely many zeros in $\mathbb{C}$, hence in any relatively compact set, unless it has none there at all — the same finiteness argument applies verbatim).*

Proof. Identical to the proof of Theorem `thm:nonDfinite` in paper2a: apply Lemma `lem:odeposes` with Ω in place of the unit disc. A pole of F_T in Ω that is not a zero of the leading coefficient a_r is impossible, and a_r has only finitely many zeros in any relatively compact subset of Ω (identity theorem, $a_r \not\equiv 0$ holomorphic), contradicting the assumed infinitude of poles there. $\square$

This lemma has zero new mathematical content beyond paper2a’s own argument; it is stated here only to make explicit that *the entire missing step of P5 is producing hypotheses (a) and (b) for some domain Ω* — nothing else is needed once they are available. This sharpens “prove Conjecture `conj:WT-nonDfinite`” to “find any domain and any continuation of F_T into it with infinitely many poles there,” which is the same reduction paper2a already performs for U, V, G_0^T , now made explicit as the target for F_T .

5 Why the hypotheses of Lemma 1 are not available

Paper1 (§8.8, lines 2732–2774) already gives the precise reason the U, V, G_0^T route does not obviously transfer, which this note restates without weakening it:

own continuation). Unlike U, V, G_0^T , there is no known representation of F_T analogous to the Fredholm determinant of the travel operator (item (a) of §3). F_T is presently known only as a power series from finite BFS layers (`wt_growth`) plus the universality identification with U ’s layers up to depth n_T . Nothing in the repository establishes that F_T continues past its radius of convergence $1/\limsup(u_d^T)^{1/d}$ at all. Hypothesis (a) of Lemma 1 is therefore currently untestable for F_T , not merely unproved.

ics look wrong). The combinatorial source of U, V ’s infinitely many poles is the travel block, a sub-transfer with catalytic variable that is *not* damped: paper1 identifies the corresponding structure for W_T as the *carry* $f \in \mathbb{Z}[t^{\pm 1}]$ in the identification $A \sim A + 2\mu_T f$ of edge profiles giving the same group element (Corollary `cor:kernel-classification-restate`). Long division of a profile by $2\mu_T$ from the low end realizes numeration in base ζ_T , $|\zeta_T| = 1$. Because the base does not contract ($|\zeta_T| = 1$, as opposed to a rational base $q/p > 1$ that contracts a remainder to a bounded window), nothing forces the carry state to stay in a bounded set: it performs a walk in the plane $\mathbb{Z}[\zeta_T]$, not a bounded automaton state.

and none exists). Paper1 states, explicitly labelled HEURISTIC there (line 2771, “We have not proved this, and state it only as a heuristic”): the travel-pole mechanism of U, V is believed to be collapsed by the carry mechanism, i.e. pure-travel elements remain distinct in W_T but the *magnitude sector* of the transfer producing the travel poles does not survive intact once carries are accounted for. No functional/transfer equation for F_T with the planar carry as an explicit catalytic variable is known to exist (paper1, line 2772–2773: “A proof of Conjecture `conj:WT-nonDfinite` would seem to need a transfer in which the planar carry is the catalytic variable, and no such equation is available.”). This is the mathematical crux: without such an equation there is no candidate route to either hypothesis (a) or (b) of Lemma 1 for F_T .

Closing P5 unconditionally therefore requires, at minimum, either (i) an explicit meromorphic continuation of F_T together with a proof of infinitely many poles inside its domain (the direct route, blocked by Gaps 1–3), or (ii) some entirely different obstruction to D -finiteness not based on pole confinement — e.g. a genuinely unbounded-state automaticity/numeration argument transferred from Akiyama–Frougny–Sakarovitch (§6), which would need to be developed from scratch, since their theorems are stated for contracting (rational, > 1) bases and $|\zeta_T| = 1$ is exactly the boundary case they do not cover.

6 The FKS / Akiyama–Frougny analogy, read precisely

Paper1 (line 2741–2750) draws the analogy this note is asked to examine. Restated precisely, with the literature flagged (§10):

- Freden–Knudson–Schofield (cited in paper1’s bibliography as `bibitem{FKS2011}`), “Growth in Baumslag–Solitar groups” or similarly titled — **FKS here stands for the surname initials of these three authors, Freden, Knudson, Schofield; it is not an acronym for Furstenberg or Flajolet–Kurshan–Steele**, contrary to one plausible guess raised in the task prompt) computed the growth of the horocyclic subgroup of $BS(p, q)$ and proved it rational when $p \mid q$.
- For $p \nmid q$ (first case $BS(2, 3)$): no geodesic combing of that subgroup is context-free, and the growth series is conjectured (not proved, per paper1) irrational; the underlying arithmetic is numeration in the rational base q/p , whose integer expansions form a non-regular language, citing Akiyama–Frougny–Sakarovitch (`bibitem{AFS2008}` in paper1, “Powers of rationals modulo 1 and rational base number systems” or similarly titled).
- Paper1 explicitly calls Conjecture `conj:WT-nonDfinite` “the unit-circle counterpart of that situation, and ... at least as hard” (line 2749–2750). The analogy is therefore stated in paper1 itself as a difficulty comparison, not as a proof strategy with a known target theorem to invoke: the $p \nmid q$ non-rationality of $BS(p, q)$ ’s horocyclic growth series is *itself an open problem* in the cited literature (per paper1’s own account), so P5’s “next step” points at a technique whose home instance is unsolved.

Consequence for P5. The suggested approach (“bounded-state structure / rational-base numeration”) is exactly the numeration-in-base- ζ_T picture of Gap 2 above, and paper1 already tried it as far as the repository’s tools allow: it looked for a bounded carry window (which would be the W_T analogue of a contracting rational base giving a finite automaton) and found none, in the numerically sampled range (below).

7 The carry-width numerics, re-read (not regenerated)

output,” not re-run) `code/zeta_probe/wt_growth/src/bin/wt_carry.rs` computes, for $(c, e) = (2, 1)$ and $d \leq 32$, the exact optimal carry width (min-plus DP over carries f , exact `i128` arithmetic in $\mathbb{Z}[1/c][\zeta]$) needed to realize the W_T -geodesic length of every element of the universal ball of radius d . Paper1 (line 2761–2763) reports this width as $\lfloor (d - 12)/2 \rfloor$ for all $d \leq 32$ — linear growth, with no plateau, in the sampled range. This is exactly the opposite of the boundedness a finite-state/bounded-carry automaton picture (the P5 “next step”) would need, in the one shape and depth range where it has been checked. Paper1 itself labels this “verified, not proved” (line 2764–2765 and again at line 23 of `wt_growth/README.md`: “The carry-window law is verified, not proved.”). I did not re-run `wt_carry` for this note (no new computation was judged necessary or worth the disk/CPU budget for a result already on file and already correctly labelled); I only re-read `code/zeta_probe/wt_growth/README.md` and the cited line range of paper1.tex to confirm the claim and its status label match.

- No sanity Rust/Arb script is added by this note. A genuinely new contribution here would be running `wt_carry` at additional (c, e) shapes and larger d to test whether the linear law $\lfloor (d - 12)/2 \rfloor$ is shape-independent or an artifact of $(c, e) = (2, 1)$; this was judged out of scope

for this reduction note (it would extend, not close, the existing VERIFIED claim) and is recorded as the first concrete next action in §9.

8 Open gap / weakest points, stated explicitly

1. The single mathematical gap that would close P5 by the pole-confinement route is exactly Gap 1 of §5: *no domain Ω and no meromorphic continuation of F_T into Ω is known*. Lemma 1 (this note) shows this is the *only* gap for that route — once (a) and (b) are supplied, non- D -finiteness follows by the identical (already-proved) argument as for U, V, G_0^T .
2. The weakest point of the existing repository evidence is Theorem `thm:WT-lowcomplexity` being read as more than it is: it is a finite-order exclusion, explicitly calibrated against a false positive (BS(1, 2), Remark `rem:BS-calibration`) that a search of comparable size would have missed. It gives no asymptotic information and cannot, by itself, ever prove Conjecture `conj:WT-nonDfinite` (a conjecture about all orders, not any finite box).
3. The carry-width linear growth (§7) is VERIFIED at one shape, one depth range ($d \leq 32$, $(c, e) = (2, 1)$), and paper1 itself does not claim it generalizes; it is not even a HEURISTIC argument for non- D -finiteness by itself, only evidence against the one concrete route (bounded carry automaton) that P5’s own suggested next step names.
4. The FKS/Akiyama–Frougny analogy (§6) is, on inspection, an analogy to an *open* problem (BS(p, q), $p \nmid q$, non-rationality), not a solved one whose method could be imported. This considerably weakens “Bounded-state structure / Rational-base numeration” as a stated next step: even a full transfer of the Akiyama–Frougny apparatus to W_T would, at best, reduce P5 to the currently-open BS(p, q) problem’s own difficulty, not solve it.
5. This note performed no new computation and no new proof beyond Lemma 1 (itself an immediate corollary of an existing lemma). It should not be read as progress on the conjecture, only as a precise map of where the difficulty is and an explicit, minimal target (Gap 1) for future work.

9 Concrete next actions (not carried out here)

1. Run `wt_carry` at additional shapes ($(c, e) = (5, -6)$ i.e. (1, 2), (53, -90) i.e. (2, 7), and one or two more laboratory shapes) and larger d (subject to the RSS/wall caps in `wt_growth/lab.sh`, `lab2.sh`, i.e. $\leq 3000\text{MB}$, $\leq 285\text{s}$ per run under `run_cap.sh`) to see whether the carry width’s linear law $\lfloor (d - 12)/2 \rfloor$ is shape-universal or shape-dependent. A shape-dependent law would be interesting; a universal one would still not prove non- D -finiteness (Gap 3), but would strengthen the heuristic.
2. Attempt to write *any* functional equation for F_T or for the toy-group growth series $\sum u_d^{\text{toy}} x^d$ with the planar carry as an explicit catalytic (extra) variable, even a conjectural one to be verified only numerically at first — this is the concrete missing object named at Gap 3, and the toy groups (already computed exactly, not modularly, in `code/zeta_probe/zeta_toy_growth/`) are the more tractable target since their growth series are known exactly rather than only through a universality bound.

3. If a transfer equation is found, check whether it exhibits a Fredholm-determinant structure whose zero set can be analyzed the way `thm:travelres` analyses G_0^T ; if so, Lemma 1 of this note applies immediately and would close P5.

10 Literature relied on — memory flags

- **Independently checked in this session:** every `paper1.tex` and `paper2a.tex` line/label cited above was read directly from the files in this repository (`paper/journal/paper1.tex`, `paper/journal/paper2a.tex`) during this session; the `wt.growth` README and `wt_carry.rs` source comments were likewise read directly.
- **Cited from paper1’s own bibliography, not independently verified against the primary source in this session** (i.e. I read paper1’s citation of them, but did not fetch or check Freden–Knudson–Schofield or Akiyama–Frougny–Sakarovitch’s papers themselves): the precise theorem statements attributed to FKS2011 and AFS2008 above are paper1’s paraphrase, taken at face value. The identification of “FKS” with Freden–Knudson–Schofield rather than some other expansion is read off paper1’s `\bibitem{FKS2011}` entry (`paper/journal/paper1.tex`, line ~5352: “E. M. Freden, T. Knudson and J. Schofield, *Growth in ...*”), which *was* read directly in this session, so this identification itself is independently checked, not a memory guess.
- **From the assistant’s training-data memory, not checked against any source in this session, and flagged as such:** (i) the general fact that Baumslag–Solitar groups $BS(p, q)$ with $p \mid q$ have rational growth series and $p \nmid q$ is open/conjectured non-rational — this matches paper1’s own statement and is not contradicted by it, but I did not independently verify it against Collins–Edjvet–Gill or Freden–Knudson–Schofield directly; (ii) any general background claim about the Akiyama–Frougny–Sakarovitch rational-base numeration system being a non-regular language and the associated automaticity theory — I recall this area exists and roughly what it says, but did not re-derive or re-check any specific theorem statement from it beyond what paper1 already states; this note’s Gap 2/Gap 3 analysis does not depend on the precise form of any AFS theorem, only on the structural observation (independently verifiable from the $(c, e) = (2, 1)$ carry data above) that $|\zeta_T| = 1$ gives a non-contracting base, which is paper1’s own point, not a citation-dependent one.
- **Stanley, *Enumerative Combinatorics* vol. II, §6.4** (equivalence of D -finite series and P-recursive coefficients): cited exactly as paper1 and paper2a cite it; not independently re-derived or checked against the book in this session.